

2005



Mathematics

Extension 2

General Instructions

- Start each question on a new page.
- Attempt questions 1, 2 and 3.
- All questions are of equal value.
- Marks for each question are shown.
- All necessary working should be shown in every question.
- Board-approved calculators may be used.
- A table of standard integrals is provided.

Total marks – 75

- Working time: 1½ hours
- Reading time: 5 minutes
- Write using blue or black pen.
- Do curve sketching questions on the number planes provided.

Question 1 – 25 marks – (Start a new page)

Marks

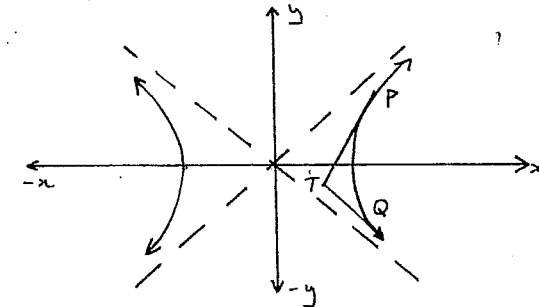
a) The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(i) Show that the equation of the normal to E at P is $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$ 3

(ii) This normal cuts the x -axis at M and the y -axis at N . Find M and N . 2

(iii) Prove that $\frac{PM}{PN} = \frac{b^2}{a^2}$ 3

b) The points $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on the hyperbola $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The point $T(x_0, y_0)$ is placed such that tangents can be drawn from T to H with points of contact P and Q as shown.



(i) Prove that the tangents to H at P and Q are 4

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \quad \text{and} \quad \frac{xx_2}{a^2} - \frac{yy_2}{b^2} = 1 \quad \text{respectively}$$

(ii) Hence show that the equation of the chord PQ is:

$$\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1 \quad 3$$

(iii) Hence show that if PQ is a focal chord then T must lie on a directrix. 2

Question 1 (cont'd)

Marks

- c) The distinct points $P\left(3p, \frac{3}{p}\right)$ and $Q\left(3q, \frac{3}{q}\right)$ (with both $p > 0$ and $q > 0$) lie

on the rectangular hyperbola $R: \begin{cases} x = 3t \\ y = \frac{3}{t} \end{cases}$

- (i) Show that the gradient of PQ is $-\frac{1}{pq}$ 1

- (ii) Show that the equation of the chord PQ is $x + pqy = 3(p + q)$ 2

- (iii) If PQ always passes through $(0, 4)$ show that the mid-point M of PQ always lies on the line $y = 2$ 3

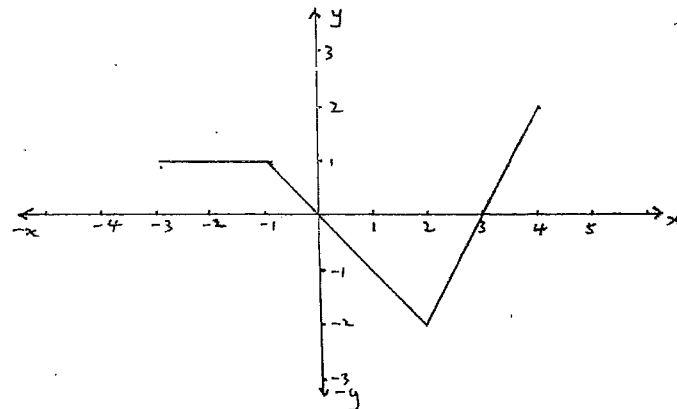
- (iv) Is it possible for M to lie anywhere on the line $y = 2$? Your answer should include any further appropriate conditions. 2

Question 2 – (25 marks) – (Start a new page)

Marks

Do all sketches on the number planes provided with the examination paper.

a)



The diagram is a sketch of the function $y = f(x)$, $-3 \leq x \leq 4$.

10

On separate diagrams (see number planes provided), draw neat sketches of:

(i) $y = |f(x)|$

(ii) $y = -f(x)$

(iii) $y = f(-x)$

(iv) $y = f(|x|)$

(v) $y = \sqrt{f(x)}$

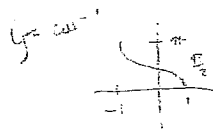
(vi) $y = \frac{1}{f(x)}$

(vii) $y = [f(x)]^2$

Question 2 (continued)

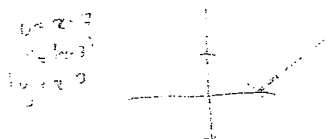
Marks

- b) (i) Sketch the curve $y = \cos^{-1}(2x)$



3

- (ii) Hence, sketch $|y| = \cos^{-1}(2x)$



2

- c) Consider the function: $y = \frac{1+x^2}{1-x^2}$

- (i) What are the equations of all its asymptotes? Give full reasons.

3

- (ii) Without using calculus, sketch $y = \frac{1+x^2}{1-x^2}$. Show all working.

3

- (iii) Hence, sketch:

(α) $y = \sqrt{\frac{1+x^2}{1-x^2}}$

For (α), show the relationship of the curve to the original function.

2

(β) $y^2 = \frac{1+x^2}{1-x^2}$

2

Question 3 – (25 marks) – (Start a new page)

Marks

- a) The ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has foci $(\pm\sqrt{7}, 0)$ and directrices $x = \pm\frac{16}{\sqrt{7}}$

- (i) Show that the eccentricity e has the value $\frac{\sqrt{7}}{4}$

3

- (ii) Show that $a = 4$ and $b = 3$

2

- (iii) Sketch E clearly showing all essential features.

2

- b) The point $P(x, y)$ lies on the hyperbola $H: \frac{x^2}{9} - \frac{y^2}{16} = 1$

H has foci S and S' .

- (i) Show that $|PS - PS'| = 6$

1

- (ii) By considering the behaviour of H for $|x|$ very large, find the asymptotes of H .

2

- (iii) Find the eccentricity e

2

Question 3 (continued)

Marks

As with Question 2, for Parts c), d) and e) of Question 3, use the number planes provided with the examination paper.

c) (i) Is the function $y = x^2 \sin x$ odd, even or neither? Give reasons. 1

(ii) Sketch the curve $y = x^2 \sin x$ for $-2\pi \leq x \leq 2\pi$, showing all important features. 2

d) (i) Sketch the curve $y = |\tan x|$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ 2

(ii) (α) Is the curve differentiable at the origin? 1

(β) Explain, using calculus, what type of point the origin is. 2

e) On separate number planes, sketch

(i) $u = \sin x$ ($0 \leq x \leq 2\pi$) 1

(ii) $y = \ln u$ 1

Hence, sketch

(iii) $y = \ln(\sin x)$ ($0 \leq x \leq 2\pi$) 3

Year 12
 Mid H.S.C. Ext. 2
 2005
 SOLUTIONS

Q1/

a) $x = a \cos \theta \rightarrow \frac{dx}{d\theta} = -a \sin \theta$
 $y = b \sin \theta$

$$\frac{dy}{d\theta} = b \cos \theta$$

$$\therefore \frac{dy}{dx} = -\frac{b \cos \theta}{a \sin \theta}$$

$$\therefore \text{slope of normal} = \frac{a \sin \theta}{b \cos \theta}$$

eqn of normal is

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

$$(x - \frac{b \sin \theta}{\sin \theta})$$

$$\text{i.e. } \frac{by}{\sin \theta} - b^2 = \frac{ax}{\cos \theta} - a^2$$

$$\therefore \frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2 \quad (3)$$

(ii) at M, $y = 0$
 $\frac{ax}{\cos \theta} = a^2 - b^2$

$$x = \frac{(a^2 - b^2) \cos \theta}{a}$$

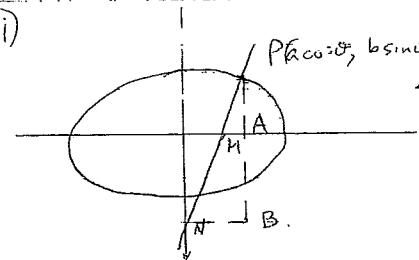
$$\therefore M \text{ is } \left(\frac{(a^2 - b^2) \cos \theta}{a}, 0 \right) \quad (2)$$

at N, $x = 0$

$$\therefore -\frac{by}{\sin \theta} = a^2 - b^2$$

$$\therefore y = \frac{(b^2 - a^2) \sin \theta}{b}$$

(ii)



$\triangle PMA \parallel \triangle PNB$. (equiangular)

$$\begin{aligned} \therefore \frac{PM}{PN} &= \frac{PA}{PB} \\ &= \frac{b \sin \theta}{b \sin \theta + \frac{(a^2 - b^2) \sin \theta}{b}} \\ &= \frac{b}{b + \frac{a^2 - b^2}{b}} \\ &= \frac{b^2}{a^2} \end{aligned}$$

(3)

b) (i) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\therefore \frac{d}{dx} \left(\frac{x^2}{a^2} - \frac{y^2}{b^2} \right) = 0$$

$$\text{i.e. } \frac{2x}{a^2} - \frac{2yy'}{b^2} = 0$$

$$\therefore y' = \frac{b^2 x}{a^2 y}$$

$$\therefore \text{at P, } y' = \frac{b^2 x_1}{a^2 y_1}$$

$\therefore$ eqn of tangent at P:

$$y - y_1 = \frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$(x - \frac{y_1}{b^2})$$

$$\frac{yy_1}{b^2} - \frac{y_1^2}{b^2} = \frac{xx_1}{a^2} - \frac{x_1^2}{a^2}$$

$$\text{or } xx_1 - yy_1 = x_1^2 - y_1^2 = 1$$

(since (x_1, y_1) lies on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$)

Similarly, tangent at Q is $\frac{x(x_2)}{a^2} - \frac{y(y_2)}{b^2} = 1$. (4)

(ii) T lies on tangent at P & at Q

$$\therefore \frac{x_0 x_1}{a^2} - \frac{y_0 y_1}{b^2} = 1 \quad (1)$$

$$\text{and } \frac{x_0 x_2}{a^2} - \frac{y_0 y_2}{b^2} = 1 \quad (2)$$

By referring to (1), (2) we deduce that $P(x_1, y_1)$ and $Q(x_2, y_2)$ both lie

$$\text{on } \frac{x x_0}{a^2} - \frac{y y_0}{b^2} = 1 \quad (3) \quad (3)$$

$\therefore$ (3) is eqn of PQ (3 is linear)

(iii) $(ae, 0)$ lies on PQ

$$\therefore (3) \Rightarrow \frac{ae x_0}{a^2} = 1. \quad (2)$$

$$\therefore x_0 = \frac{a}{e}$$

$\therefore$ T must lie on directrix $x = \frac{a}{e}$

$$\begin{aligned} \text{c) (i) gradient } PQ &= \frac{\frac{3}{2} - \frac{3}{p}}{\frac{3}{2} - 3p} \quad \times \frac{pq}{pq} \\ &= \frac{3p - 3q}{pq(3p - 3q)} \\ &= -\frac{1}{pq} \quad (1) \end{aligned}$$

(ii) Eqn of chord PQ is:

$$y - \frac{3}{p} = -\frac{1}{pq}(x - 3p)$$

$$\therefore pqy - 3q = -x + 3p \quad (2)$$

$$\therefore x + pqy = 3(p + q) \quad +$$

(iii) (0, 4) lies on +

$$\therefore 4pq = 3(p + q) \quad \oplus$$

$$M \text{ is } \left(\frac{3p + 3q}{2}, \frac{\frac{3}{p} + \frac{3}{q}}{2} \right)$$

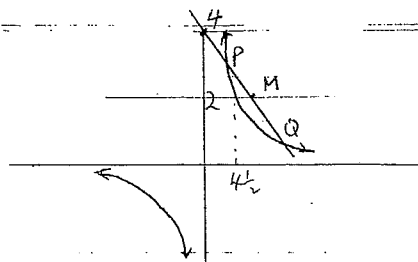
$$= \left(\frac{3}{2}(p + q), \frac{3q + 3p}{2pq} \right) \quad (3)$$

$$= \left(\frac{3}{2}(p + q), \frac{4pq}{2pq} \right) \text{ from } \oplus$$

$$= \left(\frac{3}{2}(p + q), 2 \right)$$

$\therefore$ midpoint M of PQ always lies

(iv)



(2)

Eqn of hyperbola is

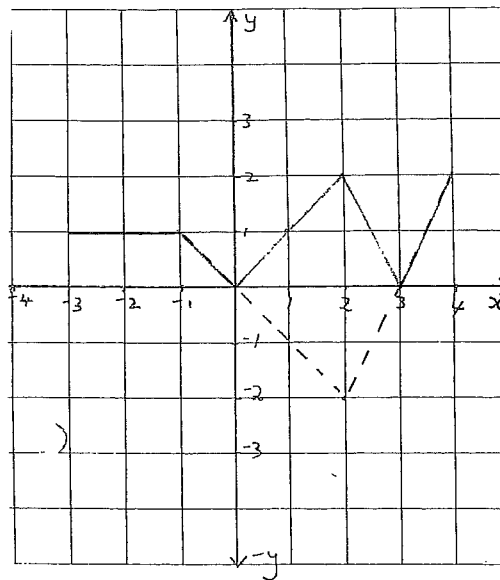
$$xy = 4$$

- when $y = 2$, $x = 4 \frac{1}{2}$

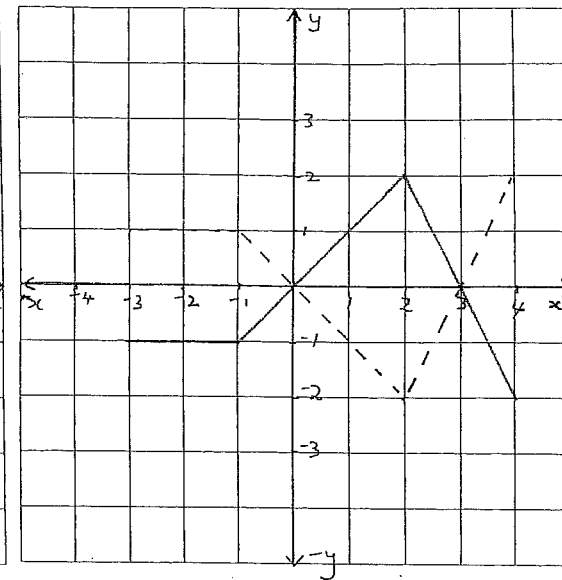
$\therefore M$ lies on $y = 2$, $x = 4 \frac{1}{2}$
(see diagram)

2 a)

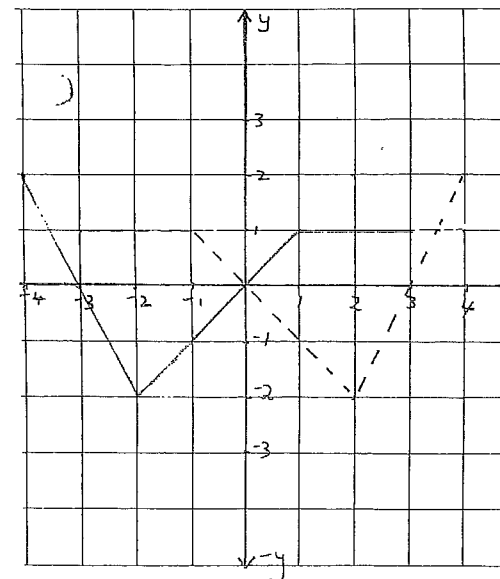
i) $y = |f(x)|$



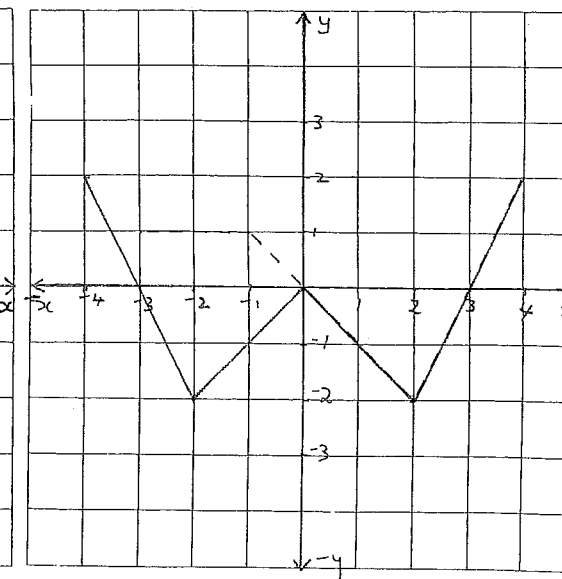
(ii) $y = -f(x)$



ii) $y = f(-x)$

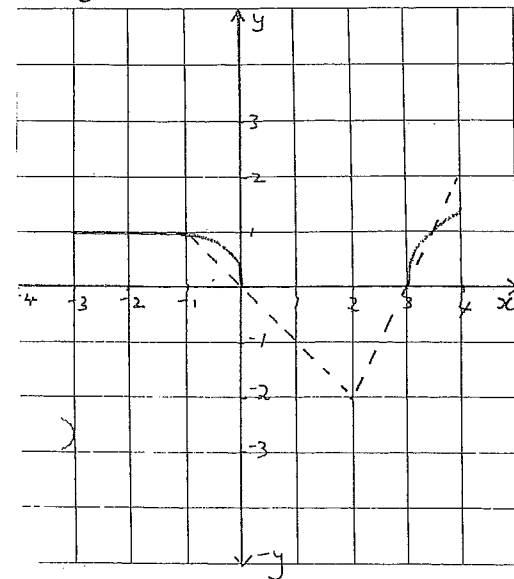


(iv) $y = f(|x|)$

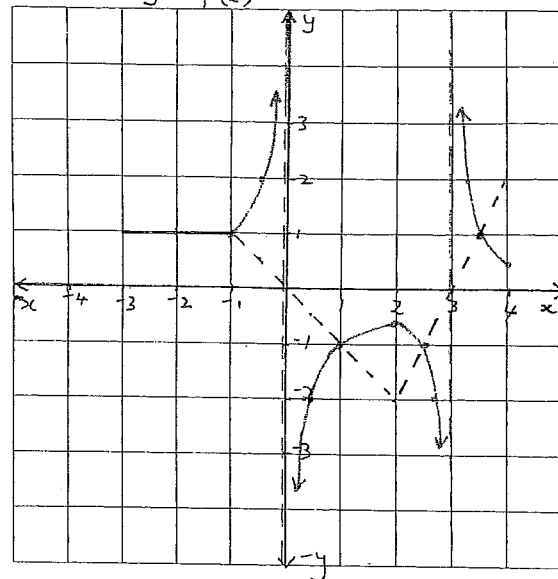


Q2 a)

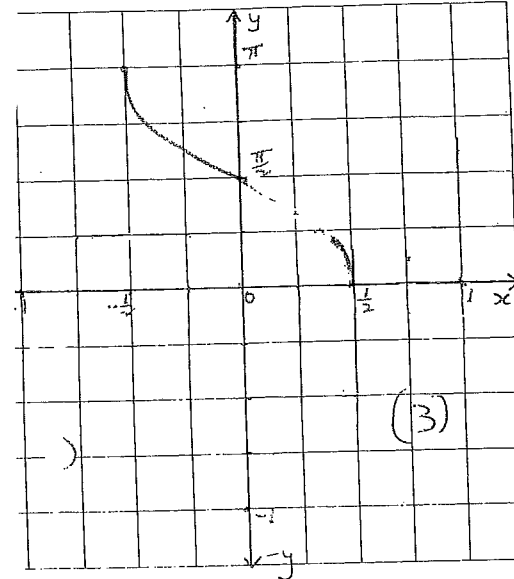
i) $y = \sqrt{f(x)}$



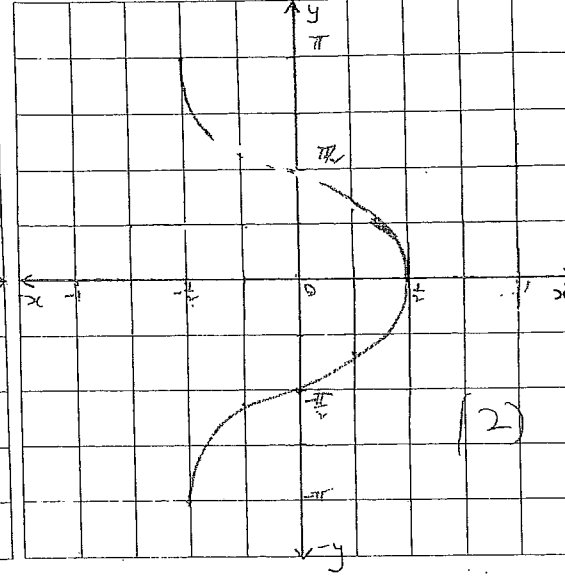
(vi) $y = \frac{1}{f(x)}$



2 b) carefully mark scale used on the axes.



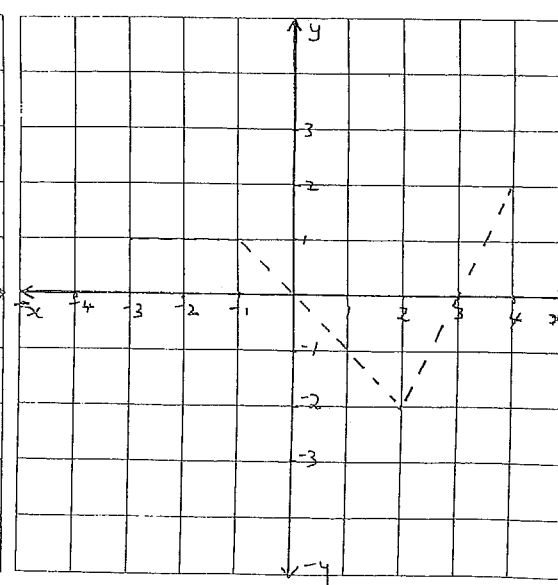
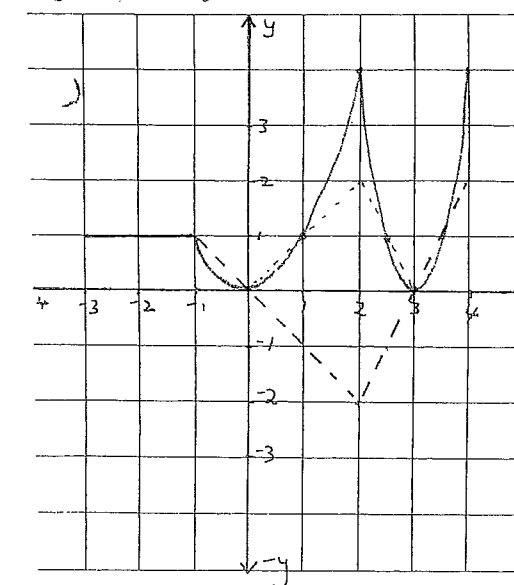
$y = \cos^{-1}(2x)$



$|y| = \cos^{-1}(2x)$

Q2 c)
(i)

i) $y = [f(x)]^2$ SPARES Please label if being used.



QUESTION 2:

(c) $y = \frac{1+x^2}{1-x^2}$

(i) vertical asymptotes: $x = -1, x = 1$

horizontal asymptotes:

$$\lim_{x \rightarrow \infty} \frac{1+x^2}{1-x^2}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2} + 1}{\frac{1}{x^2} - 1}$$

$$= -1$$

(3)

(ii) $\therefore y = -1$ is asymptote.

(iii) see graph

• as $x \rightarrow 1^+$, $\frac{1+x^2}{1-x^2} \rightarrow -\infty$

• as $x \rightarrow 1^-$, $\frac{1+x^2}{1-x^2} \rightarrow \infty$

• as $x \rightarrow -1^+$, $\frac{1+x^2}{1-x^2} \rightarrow \infty$

• as $x \rightarrow -1^-$, $\frac{1+x^2}{1-x^2} \rightarrow -\infty$

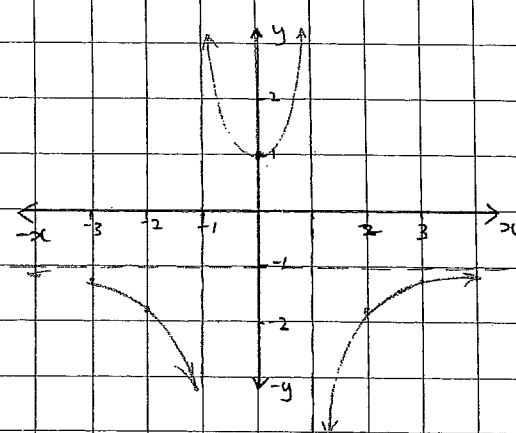
(iii) (a) see graph

(b) see graph

Q2 d) (c)

(ii)

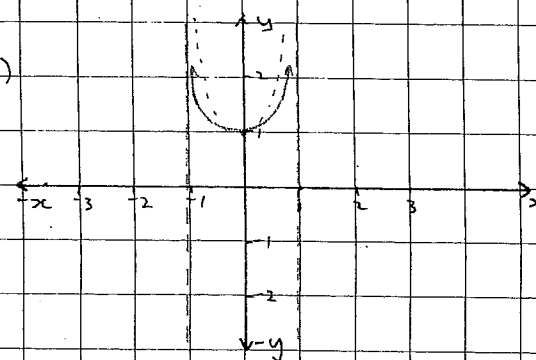
1 cm grid paper



$$y = \frac{1+x^2}{1-x^2}$$

(3)

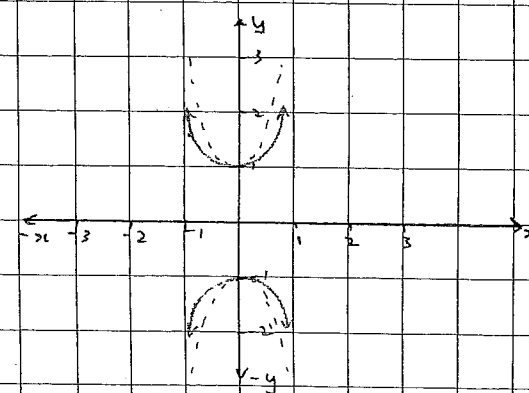
(iii) (a)



$$y = \sqrt{\frac{1+x^2}{1-x^2}}$$

(2)

(b)



$$y^2 = \frac{1+x^2}{1-x^2}$$

(2)

3
a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ foci $(\pm \sqrt{7}, 0)$

directrices $x = \pm \frac{16}{\sqrt{7}}$

$ae = \sqrt{7}$ ①

$\frac{a}{e} = \frac{16}{\sqrt{7}}$ ②

$\therefore a = \frac{16e}{\sqrt{7}}$

sub into ① $\Rightarrow \frac{16e^2}{\sqrt{7}} = \sqrt{7}$

$\therefore e^2 = \frac{7}{16}$

$e = \frac{\sqrt{7}}{4}$

(3)

(ii) sub $e = \frac{\sqrt{7}}{4}$ into ① $\Rightarrow \therefore a = 4$

$b^2 = a^2(1 - e^2)$ for an ellipse

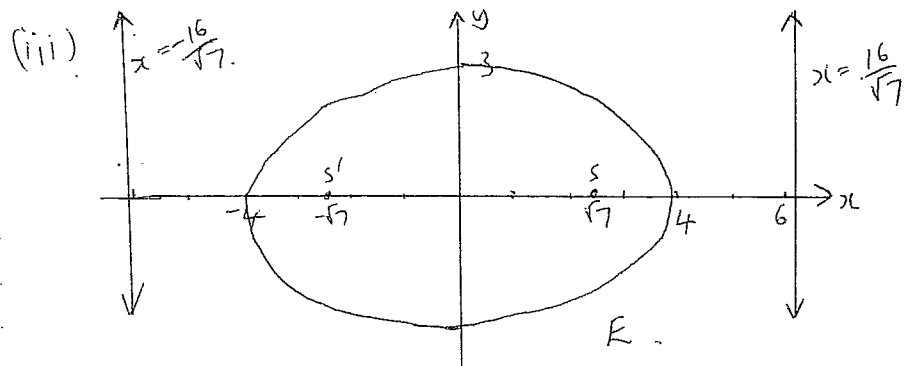
$\therefore b^2 = 16(1 - e^2)$

$= 16\left(1 - \frac{7}{16}\right)$

$= 9$

$\therefore b = 3$

(2)



(2)

b) (i) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

$\therefore a = 3, b = 4$

$PS = e PM$

$PS' = e PM'$

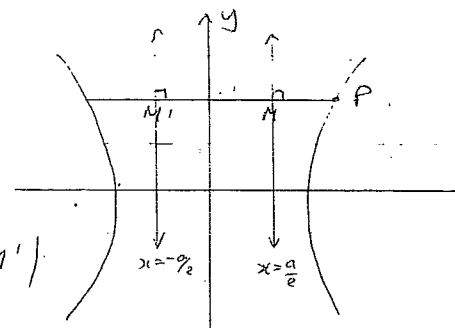
$\therefore |PS - PS'| = e |PM - PM'|$

$= e \left| \frac{2a}{e} \right|$

$= 2a$

$= 6$

(1)



(ii) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

$\therefore \frac{y^2}{16} = \frac{x^2}{9} - 1$

$y^2 = 16\left(\frac{x^2}{9} - 1\right)$

$y = \pm \sqrt{\frac{16x^2}{9} - 16} = \pm \sqrt{\frac{16x^2}{9}\left(1 - \frac{9}{x^2}\right)}$

$= \pm \frac{4x}{3} \sqrt{1 - \frac{9}{x^2}}$

(2)

as $x \rightarrow \pm\infty, y \rightarrow \pm \frac{4x}{3}$

$\therefore$ asymptotes are $y = \pm \frac{4x}{3}$

(iii) $b^2 = a^2(e^2 - 1)$

$\therefore 16 = 9(e^2 - 1)$

$e^2 = \frac{25}{9}$

$e = \frac{5}{3}$

(2)

QUESTION 3

(c) (i) $f(x) = x^2 \sin x$
 $f(-x) = (-x)^2 \sin(-x)$
 $= -x^2 \sin x \quad (1)$
 $= -f(x)$

$\therefore f(x)$ is an odd function.

(ii) see graph

(d) (i) see graph

(ii) (a) NO (1)

(b) For $x > 0$, $|\tan x| = \tan x$

If $f(x) = |\tan x| = \tan x$ for $x > 0$
 $\therefore f'(x) = \sec^2 x$

as $x \rightarrow 0^+$, $f'(x) \rightarrow 1$ ——— (1)

For $x < 0$, $|\tan x| = -\tan x$

$\therefore f(x) = |\tan x|$
 $= -\tan x$ for $x < 0$

$f'(x) = -\sec^2 x$ (2)

as $x \rightarrow 0^-$, $f'(x) \rightarrow -1$ ——— (2)

Since the two limits are not equal,

$f'(x)$ is not differentiable at $x=0$

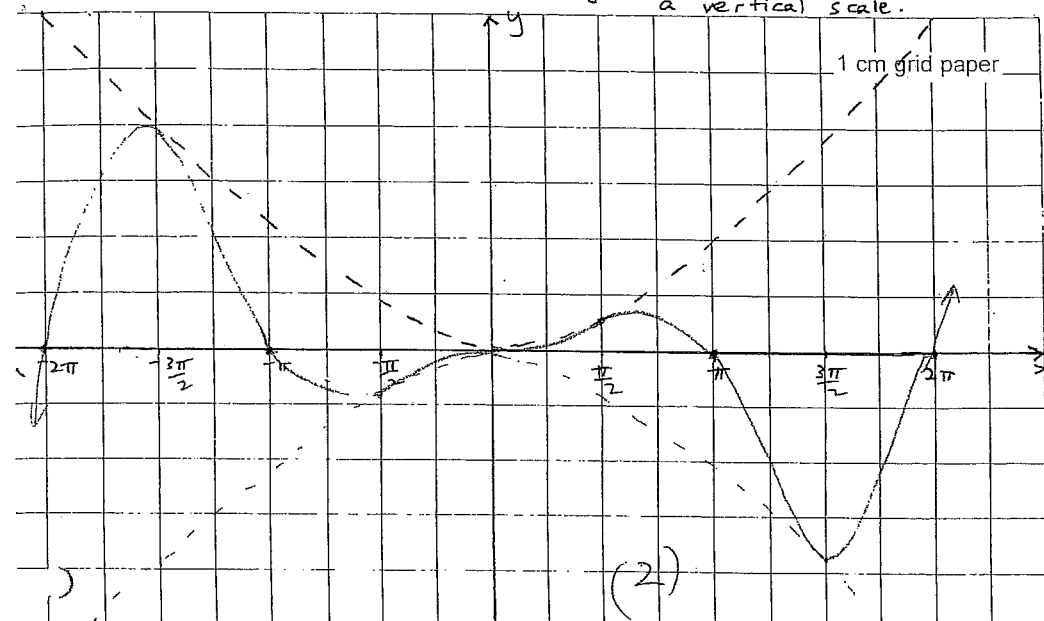
and hence the origin is a cusp.

(e) (i) see graph

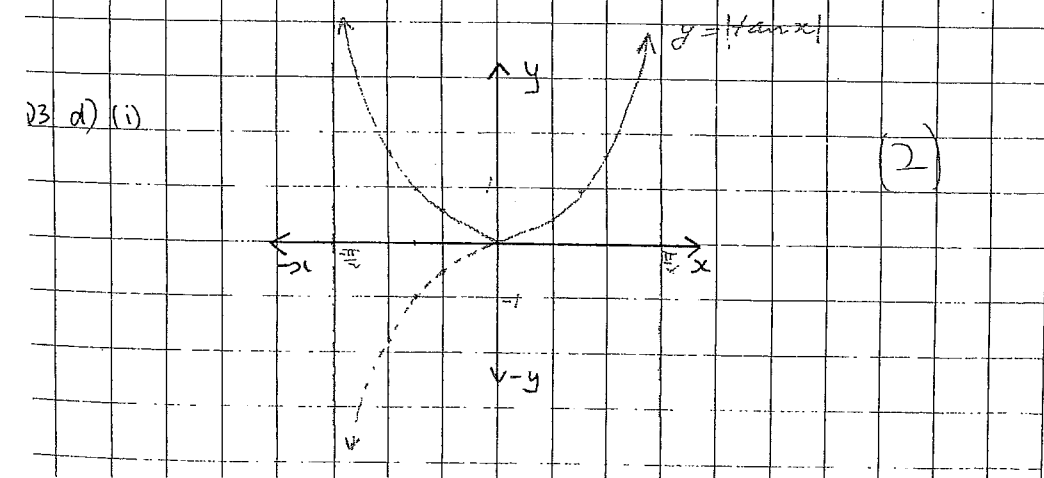
(ii) see graph

(iii) see graph

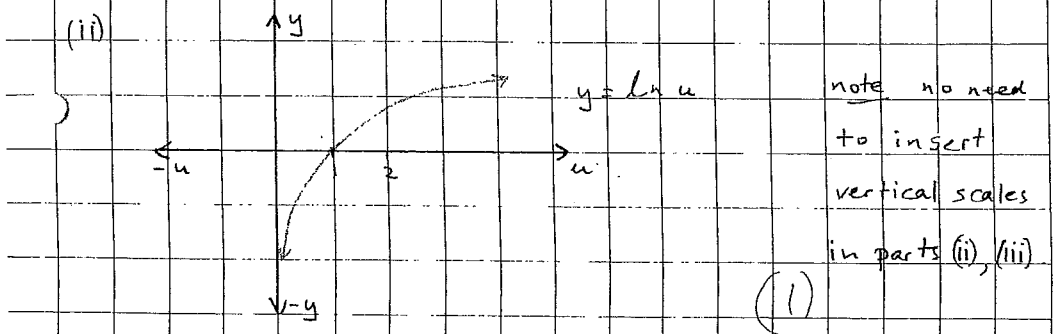
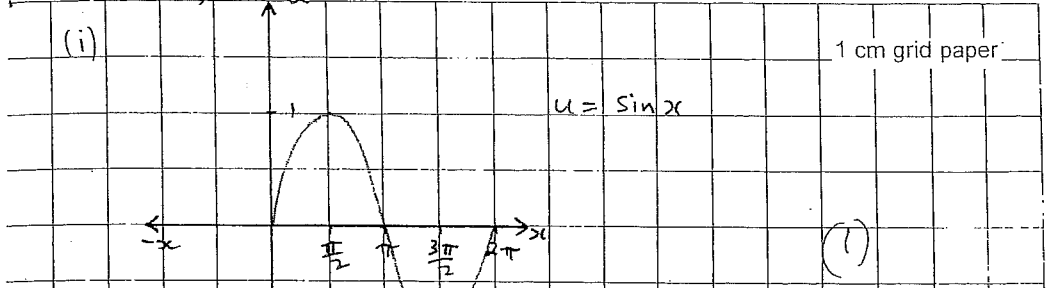
Q3 c) (ii) The dotted curve is $y=x$. There is no need to show a vertical scale.



WORKING SPACE FOR Q3 d) (i)



Q3 e)



note no need to insert vertical scales in parts (ii), (iii)

